Distinguishing Ignorance from Error in LLM Hallucinations

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arXiv:2410.22071v2 [cs.CL] 18 Feb 2025

Abstract

Large language models (LLMs) are susceptible to hallucinations—factually incorrect outputs—leading to a large body of work on detecting and mitigating such cases. We argue that it is important to distinguish between two types of hallucinations: ones where the model does not hold the correct answer in its parameters, which we term HK^-, and ones where the model answers incorrectly despite having the required knowledge, termed HK^+. We first find that HK^+ hallucinations are prevalent and occur across models and datasets. Then, we demonstrate that distinguishing between these two cases is beneficial for mitigating hallucinations. Importantly, we show that different models hallucinate on different examples, which motivates constructing model-specific hallucination datasets for training detectors. Overall, our findings draw attention to classifying types of hallucinations and provide means to handle them more effectively. [1]

1 Introduction

LLMs are prone to generate outputs that lack grounding in the model's input or in real-world facts, as well as outputs that may be inconsistent with earlier generations within the same session (Ji et al., 2023; Sharma et al., 2023; Kalai and Vempala, 2023). These issues, collectively known as *hallucinations*, are critical to address due to their impact on LLM reliability.

Numerous studies have focused on the detection and mitigation of hallucinations (e.g. Li et al., 2023; Zhang et al., 2024a; Marks and Tegmark, 2023; Chen et al., 2024; CH-Wang et al., 2023). However, existing work does not fully investigate the different causes of hallucinations, often blending two distinct types:

HK^- The model lacks the required information for a correct answer, leading it to hallucinate.

HK^+ Although the model generates correct answers under certain prompts, it still produces incorrect responses in different but *similar* prompt settings.

These types represent fundamentally different problems, requiring different solutions: When a model lacks knowledge, it should consult external sources or abstain, but when a model inherently holds the knowledge in its parameters, it may be possible to prompt it or intervene in its computation to obtain the correct answer.

In this work, we demonstrate the prevalence of HK^+, showing that it occurs in up to 24% of cases (Section 3.3). Importantly, we observe that HK^+ errors manifest even when using mild prompts that neither contain misinformation nor induce hallucination-prone scenarios. Thus, these hallucinations are not merely a consequence of flawed human prompts but can arise independently.

To study HK^+, HK^-, and factually-correct examples, we developed an automatic approach to collect **W**rong **A**nswers despite having **C**orrect **K**nowledge (WACK). Our approach (Figure 1) constructs a *model-specific* hallucination dataset that labeled hallucination types —HK^- and HK^+— by introducing novel ways of constructing HK^+. The process assesses the target model's knowledge by inspecting the number of correct responses in a neutral prompt out of multiple output samples for questions with curated known answers. For questions in which the model indicates knowledge, WACK explores altered prompt settings, which induce hallucinations in scenarios that mimic typical interactions with an LLM. These prompts reveal HK^+ examples, in which the model answers incorrectly even though it holds the knowledge for a correct answer.

After establishing the prevalence of HK^+ and creating a framework to investigate the hallucination cases, we provide a proof of concept showing that mitigation can benefit from distinguishing HK^+ from HK^- (Section 4.1). Next, we investigate how different types of hallucinations (HK^- and HK^+) are represented within models. We do this by training linear classifiers on each model's internal states, a common technique in hallucination detection research (CH-Wang et al., 2023; Azaria and Mitchell, 2023). While prior work focuses on detecting whether any hallucination occurs, we demonstrate that simple classifiers can distinguish between the two hallucination types to some extent (Section 4.2). To further demonstrate the utility of our setup for investigating HK^+, we show that HK^+ generalizes across different prompt settings; a classifier trained on examples from one prompting setting is able to predict hallucinations in another setting (Section 5.2).

Our last contribution shows that the WACK datasets

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[1]Code and datasets at https://github.com/technion-cs-nlp/hallucination-mitigation.

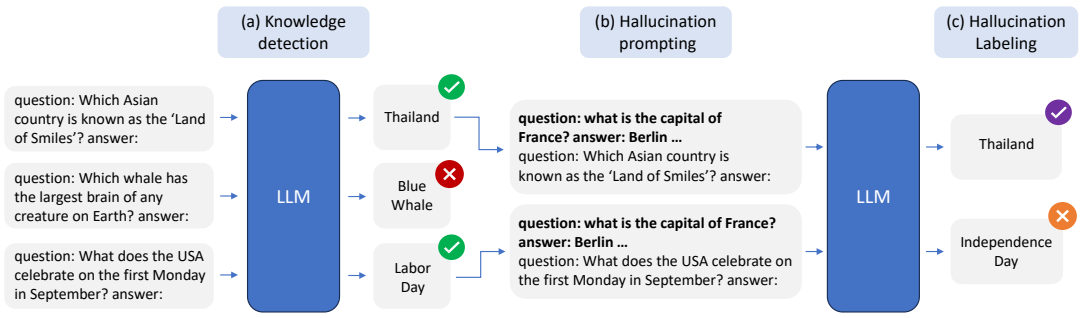


Figure 1: **WACK Setup:** (a) The first step in our process involves detecting whether the model knows the correct answer. If the model does not know the correct answer, the example is labeled as hallucination caused by not knowing (HK^-) If the model knows the correct answer, we proceed to the next stage. (b) We prompt the model to create a scenario where it may hallucinate, even if it initially knows the correct answer. Here we show a snowballing prompt. (c) Under the new setting, if the model generates the correct answer, the example is labeled as factually-correct; otherwise, it is labeled as a hallucination despite knowing (HK^+).

differ between models, highlighting the importance of using model-specific datasets that account for the unique knowledge and hallucination patterns of each model (Section 5.1). As a result, we demonstrate that model-specific datasets are more effective for HK^+ detection than datasets that ignore the HK^-/HK^+ distinction (Sections 6.1 and 6.2). Using this novel framework and its results, we can improve the detection and mitigation of hallucinations.

Our main contributions are: (I) Indicate the prevalence of HK^+ hallucinations by showing their consistency under varying prompts, including supposedly factually-encouraging prompts. (II) Demonstrating that mitigation can benefit from the separation of the two types of hallucination. (III) Proposing WACK, an approach for constructing model-specific datasets that include factually correct and hallucination examples both due to lack of knowledge (HK^-) and despite having knowledge (HK^+). We will release the datasets we constructed. (IV) Showing the importance of model-specific datasets for HK^+ detection and the ability for preemptive detection.

2 Related Work

Hallucinations. One approach for detecting hallucinations treats the model as a black box, posing questions or sampling its outputs (Gekhman et al., 2023; Pacchiardi et al., 2023; Manakul et al., 2023; Li et al., 2024a). Another line of works examines the model's hidden representations, often via a classifier (Burns et al., 2022; He et al., 2023; Rateike et al., 2023; Slobodkin et al., 2023; Azaria and Mitchell, 2023; CH-Wang et al., 2023; Yuksekgonul et al., 2023; Chen et al., 2023; Yin et al., 2024; Levinstein and Herrmann, 2024; Marks and Tegmark, 2023; Li et al., 2023). There is also ongoing research on mitigating hallucinations by steering the model towards correct answers, including prompting (Si et al., 2022; Dhuliawala et al., 2023), fine-tuning (Tian et al., 2023; Razumovskaia et al., 2023), and modifying model logits (Chuang et al., 2023; Kalai and Vempala, 2023; Li et al., 2023; Zhang et al., 2024a). These works do not distinguish between types of hallucinations, which may hinder performance of both detection and mitigation. Specifically, in Section 4.1 we show that mitigation via prompting is plausible for HK^+ but not for HK^-, and that separating between HK^+ and HK^- would result in performance improvement.

Most prior work in this area uses generic datasets. Some works explore model-specific hallucination datasets and demonstrated their importance (Azaria and Mitchell, 2023; Ji et al., 2024; Cao et al., 2023; CH-Wang et al., 2023). However, they do not distinguish between the causes of hallucinations. We show that model-specific datasets with HK^- and HK^+ typing can be beneficial for detection and mitigation, and that they are useful for preemptive detection (Section 6).

Knowledge. Several studies attempt to assess the knowledge stored in a model after it has been trained, including prompting, sampling, using paraphrases, or training to select model knowledge (Petroni et al., 2019; Jiang et al., 2020; Elazar et al., 2021; Kadavath et al., 2022; Gekhman et al., 2024; Jiang et al., 2024; Ren et al., 2023; Zhang et al., 2024b). Similarly in this work, we use greedy decoding and sampling to evaluate the model's knowledge.

A few recent studies detect cases of missing knowledge (Wen et al., 2024; Zhang et al., 2024b), similar to our HK^- hallucinations. However, they focus on using external knowledge to mitigate them while we focus on the advantage of HK^- and HK^+ typing in understanding the model's activations.

Jailbreaking. Jailbreaking refers to techniques that cause LLMs to generate unexpected or incorrect answers. For instance, Zhang et al. (2023) demonstrate the snowballing effect, where once the model outputs a wrong fact, it is more likely to generate an incorrect explanation for that fact. Additionally, research found

Model	Prompt	Response	
		Original	Truthful setting
Gemma-2-9B	question: What is the technical term for a solid figure with five plane (flat) faces? answer:	pentahedron	pentagon
Llama-3.1-8B	question: What is the national animal of Greenland? answer:	Polar bear	Arctic fox
Mistral-7B-v0.3	question: The Java Trench is part of which ocean?answer:	Indian Ocean	Pacific Ocean

Table 1: Generated answers using greedy decoding on TriviaQA, illustrating the HK^+ using Truthful setting.

that a model's answers can change due to persuasion, long conversations, fantasy settings, LLM personas, and out-of-distribution prompts (Zeng et al., 2024; Li et al., 2024b; Xu et al., 2023; Yao et al., 2023; Nardo, 2023; Joshi et al., 2023; Pacchiardi et al., 2023; Meinke et al., 2024). These studies show that prompt characteristics heavily influence output correctness, causing hallucinations even when the model knows the answer (HK^+).

Yet, they only demonstrate the phenomenon but do not explore its representation, detection, and mitigation. In this work we dive into the HK^+ phenomenon, define it properly, quantify its occurrence across models, prompts and datasets, and emphasize its importance when attempting to mitigate hallucinations.

3 WACK Framework

In this section, we outline the creation of model-specific datasets for the closed-book question answering (CBQA) setting with short answers, with a focus on generating HK^+ examples. Figure 1 provides an overview of our approach. The first step classifies examples based on the model's knowledge, labeling all instances where the model lacks knowledge as HK^-. Next, for each example where the model has the knowledge for a correct answer, we create a prompting scenario in which hallucinations could occur despite the model's knowledge (HK^+). The next subsections describe these steps.

3.1 Categorization of Knowledge

In our CBQA setting, given a question q with a known gold answer a_g, a target model generates an answer $\tilde{a}$, which may match a_g or else constitute a hallucination. Knowledge in a language model can be viewed as lying on a spectrum. We refer to the model's parametric knowledge as being at the 'low-knowledge end' when there is little to no association between a_g and q, and at the 'high-knowledge end' when this association is strong. The model is unlikely to generate the correct answer in the low-knowledge end, whereas in the high-knowledge end, it typically does.

Hallucinations at the low-knowledge end of the spectrum are somewhat expected, as the model is unlikely to generate a_g (that is, we expect $\tilde{a} \neq a_g$). However, hallucinations can occur anywhere along this spectrum, including at the high-knowledge end. Detecting the cause of hallucinations in the middle of the spectrum is more complex, and is beyond the scope of the current work.

To simplify our analysis, we focus on the two ends of the spectrum: high-knowledge and low-knowledge, which still provide a compelling overview of the two types of hallucinations. Given model M, we follow the setup of Gekhman et al. (2024) in which M generates various completions to q, and then we verify the existence of the answer a_g in the output.[2] If the model did not generate a_g in any of the generations, the example is labeled HK^-. If the model generates a_g in all attempts, the example is considered a high-knowledge scenario, to be labeled as either factually-correct or HK^+.

Our categorization of examples according to the model's knowledge may not be perfect. However, as Appendix A shows, different configurations (temperature, number of generations, etc.) produce similar labeling outcomes, which gives confidence in the process.

3.2 HK^+ Settings

To label a high-knowledge example as either factually correct or hallucination (HK^+), we design four synthetic setups that nudge the model toward producing an incorrect answer with progressively stronger settings. These setups are inspired by recent observations that various prompt modifications induce hallucinations, including perturbations (Yao et al., 2023), adding persuasive language (Xu et al., 2023; Zeng et al., 2024), and other "snowballing" effects, where one error by a model makes it generate incorrect explanations to justify the error (Zhang et al., 2023) and similarly that contextual errors can make the model generate more mistakes (Meinke et al., 2024). Notably, we observe that even completely error-free settings can produce HK^+ examples. Later in the paper, we demonstrate that one setting generalizes to another, indicating that it is valid to use these synthetic setups to investigate the general phenomenon of HK^+.

Truthful setting. This setting represents a simple prompt modification, which should encourage the model to generate truthful responses. We prompted ChatGPT (OpenAI, 2022) to generate 10 paraphrased variations of the instruction: "Generate answers that are entirely factual and precise". For each example, we included a one-shot example with a permutation of the text. The 10 paraphrased prompts are provided in Appendix B.2.

[2] Specifically, we perform one greedy generation plus five generations with a temperature of 0.5. We use a 3-shots in-context learning scenario (Brown et al., 2020), generate a maximum of 5 tokens, and look for an exact match to a_g. See Appendix A for the decision process of the hyperparams.

	TriviaQA				Natural Questions			
	Truthful	Persona	Alice-Bob	Snowballing	Truthful	Persona	Alice-Bob	Snowballing
Gemma	3.97%	4.05%	4.22%	15.92%	8.24%	8.97%	8.98%	23.52%
Llama	5.32%	4.85%	6.18%	10.58%	11.86%	12.57%	13.91%	15.69%
Mistral	4.94%	5.93%	6.38%	18.34%	9.67%	12.52%	12.32%	21.75%

Table 2: Percentage of HK^+ in TriviaQA and Natural Questions datasets with different settings. As the setting becomes more extreme (moving from left to right in each block), the HK^+ percentage increases.

Table 1 shows one example per model from the TriviaQA dataset (Joshi et al., 2017), where the model generated the correct answer using 3 random good shots (regular-few-shots), but hallucinated when prompted with this setting.

Persona setting. To create a diverse, persona-driven approach to truthful response generation, we selected 10 personas from Ge et al. (2024) and prompted ChatGPT to rewrite the following text as if each persona wrote it: "I would really like to know the answers to the following very hard questions. Could you please explain them to me?". For each example, we again incorporated a one-shot example, pairing a permutation of the text with a different persona. The 10 rewritten prompts are provided in Appendix B.2.

Alice-Bob setting. The Alice-Bob setting uses persuasion and text perturbations (typos) to simulate an error, inspired by related studies (Xu et al., 2023; Yao et al., 2023; Zeng et al., 2024). Specifically, we add the following text at the beginning of the prompt (typos are underlined), along with a one-shot example at the end:

Alice and Bob are students studying for a test on general world knowledge that requires a vast amount of understanding and knowledge. Alice is the smarter of the two. They are given four questions in total. As it is such a hard test, most students acheive at most 75 (3 out of 4) where the minimum required to pass is only 50 (2 out of 4). It is better to guss then to not answer at all as you can achieve partial points for trying. Bob and Alice only need to pass the test. Alice will ask the questions and Bob will try his best.

The persuasive aspect of the setting comes through several nuances in the text: (1) there is an implication that Bob is not smart, (2) the test is described as difficult, (3) to pass, one only needs to be correct on 2 out of the 4 questions, and (4) there is no suggestion that exceeding the minimum required score offers any advantage.

Snowballing setting. This setting illustrates how mistakes in the context can create a cascading effect that may compromise the accuracy of the model's subsequent generations. Specifically, we construct 20 false QA pairs using ChatGPT 3.5, where the false answer is semantically similar to the correct one (Appendix B.2). For instance, given the question "*Which element has the chemical symbol 'H'*", with the gold answer "*Hydrogen*", we obtained the answer "*Helium*". For each test question, we prepend k random snowballing examples in a few-shot manner, thus simulating mistakes that a user or a model might create as part of the input context.[3]

This setting is crucial to study as it reflects prompt diversity or errors, which can stem from either the user or the model's prior outputs. User errors, such as incorrect facts or language mistakes, represent real-world challenges that can lead to model hallucinations and must be addressed. Similarly, errors in the model's earlier outputs can create a snowballing effect, compounding mistakes in subsequent prompts and generations. Mitigating both user- and model-originated errors is essential for reducing hallucinations and ensuring reliable performance.

These settings differ in their degree of mildness. In the Truthful setting, the model is directly prompted to answer the question correctly (an error-free setting). The persona setting introduces additional context of story elements that may confuse the model. The Alice-Bob setting increases complexity further by including typos and implying that the answer might be incorrect. Lastly, the Snowballing setting represents the least mild scenario, where the model is explicitly prompted with incorrect answers.

3.3 HK^+ is a Broad Phenomenon

Equipped with our process for separating examples of low and high knowledge (Section 3.1) and further encouraging hallucinations for high-knowledge examples (Section 3.2), we create model-specific datasets. To align with prior work (Li et al., 2023; Zhang et al., 2024a; Chen et al., 2023), we conduct our study using two common general English CBQA datasets: TriviaQA (Joshi et al., 2017) and NaturalQuestions (Kwiatkowski et al., 2019). We experiment with three models: Mistral-7B-v0.3 (Jiang et al., 2023), Llama-3.1-8B (Dubey et al., 2024), and Gemma-2-9B (Team et al., 2024). For more details regarding the dataset construction and additional qualitative evaluation, see Appendix B.

Table 2 provides the HK^+ percentage statistics. Across all prompt settings, the models generate 4%–24% HK^+ hallucinations out of all cases of high knowledge. The Snowballing setting leads to the highest rate of HK^+ examples. Yet, **HK^+ hallucinations occur even in very mild settings** like Truthful and Persona, highlighting that despite high knowledge (strong

[3]We use $k = 3$ in the main paper. The results are consistent with other choices; Appendix K.